



The Pioneer Of Geosynthetics
S I N C E 1 9 7 2

GSE GundSeal Geosynthetic Clay Liner (Textured HDPE)

GSE GundSeal geosynthetic clay liner (GCL) is a composite liner system that consists of a high quality sodium bentonite adhered to a textured high density polyethylene (HDPE) geomembrane with a spunbonded geotextile to protect the bentonite during installation. This one product composite liner system combines the low permeability of an HDPE geomembrane with the self-seaming characteristics of bentonite clay. The intimate contact of the bentonite with the geomembrane provides the best leak protection in the industry.

Product Specifications

TESTED PROPERTY	TEST METHOD	FREQUENCY	MINIMUM AVERAGE VALUE				
FINISHED GCL PROPERTY			20 mil	30 mil	40 mil	60 mil	80 mil
Bentonite Coating ⁽¹⁾ , lb/ft² (kg/m²)	ASTM D 5993	1/40,000 ft² (1/4000 m²)	≥ 0.75 (3.66)				
Effective Hydraulic Conductivity, m/s	ASTM D 5887/E 96	periodically	≤ 4 x 10 ⁻¹⁴				
Bentonite Moisture Content	ASTM D 2216	1/40,000 ft² (1/4000 m²)	25% Typical				
GEOMEMBRANE PROPERTY ⁽²⁾							
Thickness, (minimum average) mil (mm)	ASTM D 5994	1/100,000 ft² (1/10,000 m²)	18 (0.45)	30 (0.75)	40 (1.00)	60 (1.50)	80 (2.00)
Lowest individual reading (-10%)			18 (0.45)	27 (0.69)	36 (0.91)	54 (1.40)	72 (1.80)
Density, g/cm³	ASTM D 1505	1/200,000 ft² (1/20,000 m²)	0.94	0.94	0.94	0.94	0.94
Tensile Properties							
Tensile Break Strength, lb/in (N/mm)	ASTM D 6693	1/200,000 ft² (1/20,000 m²)	30 (5)	66 (11)	75 (13)	115 (20)	155 (27)
GCL Tensile Strength ⁽³⁾ , lb/in (N/mm)	ASTM D 6768	1/200,000 ft² (1/20,000 m²)	40 (7)	63 (11)	84 (15)	130 (23)	173 (30)
Elongation at Break, %	ASTM D 6693	1/200,000 ft² (1/20,000 m²)	100	100	100	100	100
Puncture Resistance, lb (N)	ASTM D 4833	1/200,000 ft² (1/20,000 m²)	30 (133)	65 (289)	95 (422)	130 (578)	160 (711)
SODIUM BENTONITE PROPERTY							
Hydraulic Flux: Bentonite, m³/m²/sec	ASTM D 5887	periodically	≤ 1 x 10 ⁻⁸				
Hydraulic Conductivity, m/s	ASTM D 5887	periodically	≤ 5 x 10 ⁻¹¹				
Swell Index, ml/2 g	ASTM D 5890	1/60,000 lb (1/30,000 kg)	≥ 24				
Fluid Loss, ml	ASTM D 5891	1/60,000 lb (1/30,000 kg)	≤ 18				
TYPICAL ROLL DIMENSIONS							
Roll Width ⁽⁴⁾ , ft (m)			17.5 (5.3)	17.5 (5.3)	17.5 (5.3)	17.5 (5.3)	17.5 (5.3)
Roll Length ⁽⁴⁾ , ft (m)			180 (54)	180 (54)	170 (51)	170 (51)	150 (45)
Roll Area, ft² (m²)			3,150 (286)	3,150 (286)	2,975 (276)	2,975 (276)	2,625 (244)
Roll Weight, lb (kg)			3,900 (1,770)	4,100 (1,870)	4,300 (1,940)	4,600 (2,090)	4,400 (2,000)

NOTES:

- ⁽¹⁾0% moisture content.
- ⁽²⁾See specific GSE HD Textured geomembrane product data sheet for additional information.
- ⁽³⁾4 in (101 mm) wide sample, 12 in/min (305 mm/min). Values are representative of the geomembrane tensile yield strength.
- ⁽⁴⁾Roll lengths and widths have a tolerance of ± 1%.